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Measurement of Flux Density or Measurement of flux.

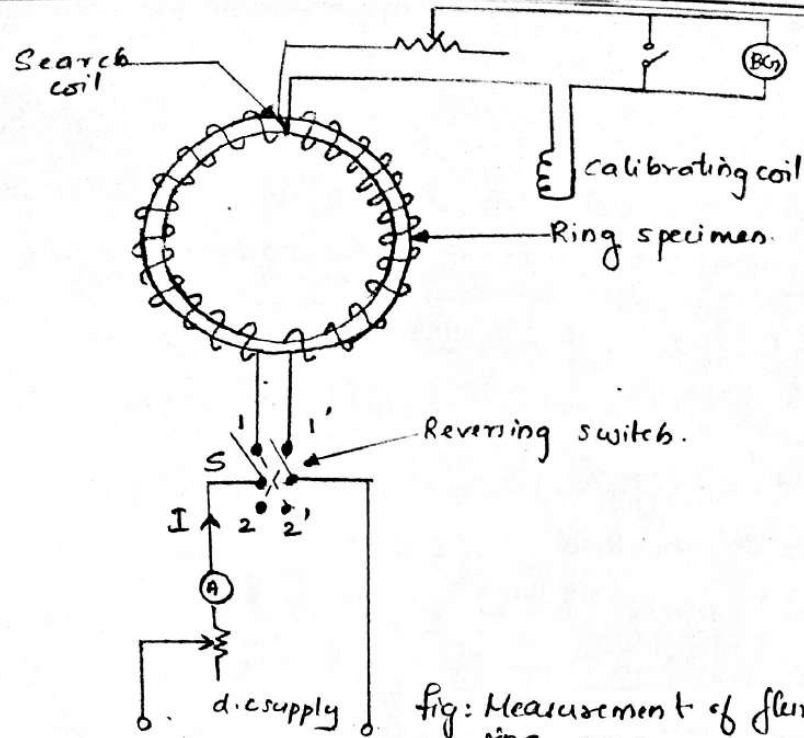


Fig: Measurement of flux density in ring specimen.

Constructional details.

- >> The flux density B inside a ring specimen can be measured by winding a search coil (B-coil) over the ring specimen.
- >> The search coil is connected to a Ballistic Galvanometer or to a flux meter.
- >> Fig. shows, the ring specimen is wound with a magnetizing winding which carries a current I . A search coil of a convenient no. of turns is wound on the specimen and connected through a resistance and calibrating coil, to a ballistic galvanometer, shown in fig.
- >> The current through the magnetizing coil is reversed & $\therefore$ the flux linkage of the search coil changes inducing an emf in it. This emf sends a current through the Ballistic galvanometer (BG) causing it to deflect.

Let ϕ = flux linking the search coil.

R = resistance of BG circuit.

N = no. of turns in the search coil.

t = time taken to reverse the flux.

$\Rightarrow \therefore$ Avg: Emf induced in the search coil $e = \frac{N d\phi}{dt} = \frac{2N\phi}{t}$ the deflection $\rightarrow$ ①

Note: Let ϕ = Initial flux in wb.

$-\phi$ = flux after reversal in wb.

Hence change in flux, $d\phi = \phi - (-\phi) = 2\phi$.

In this method $dt = t$.

$$\text{Hence } e = \frac{N d\phi}{dt} = \frac{N 2\phi}{t} = \frac{2N\phi}{t}$$

$\Rightarrow$ Avg: current through BG, $i = \frac{e}{R} = \frac{2N\phi}{tR} = \frac{2N\phi}{Rt}$ in Ampere. ②

Hence charge passing is $Q = it = \frac{2N\phi}{Rt} \times t = \frac{2N\phi}{R}$ ③

$\Rightarrow$ The deflection of the galvanometer is proportional to the charge passing through it.

$$\text{Hence ; } \theta \propto Q \text{ --- (4)}$$

$$Q = k_g \theta ; k_g \rightarrow \text{galvanometer constant} \text{ --- (5)}$$

sub (3) in (5)

$$\frac{2N\phi}{R} = k_g \theta$$

$$\phi = \frac{k_g R \theta}{2N} \text{ wb --- (6)}$$

We know that, B , flux density $B = \frac{\text{flux}}{\text{area}} = \frac{\phi}{A_s} = \frac{k_g R \theta}{2N A_s}$

$$\text{Hence } B = \frac{k_g R \theta}{2N A_s} \text{ wb/m}^2 \text{ --- (7)}$$

Thus the flux density can be measured by the deflection of the BG.

Correction for air flux.

Eqn: (7) holds good only if the assumption mentioned above is satisfied, the search coil is usually of larger area than the ring specimen and hence the flux linking with the search coil is the sum of the flux existing in the ring specimen & the flux present in the airgap between the specimen and the search coil

$\Rightarrow \therefore$ Observed value of flux = true value of flux in the specimen + flux in the airgap between the specimen and search coil

$$B' A_s = B_t A_s + \mu_0 H (A_c - A_s) \quad \text{--- (8)}$$

Hence true value of flux density

$$B_t = B' - \mu_0 H \left(\frac{A_c}{A_s} - 1 \right) \quad \text{--- (9)}$$

where B' = observed value of flux density, wb/m^2 .

B = true value of flux density in specimen, wb/m^2 .

A_s = area of cross section of specimen; m^2 .

A_c = area of cross section of coil; m^2

Measurement of magnetising force (H) or permeability

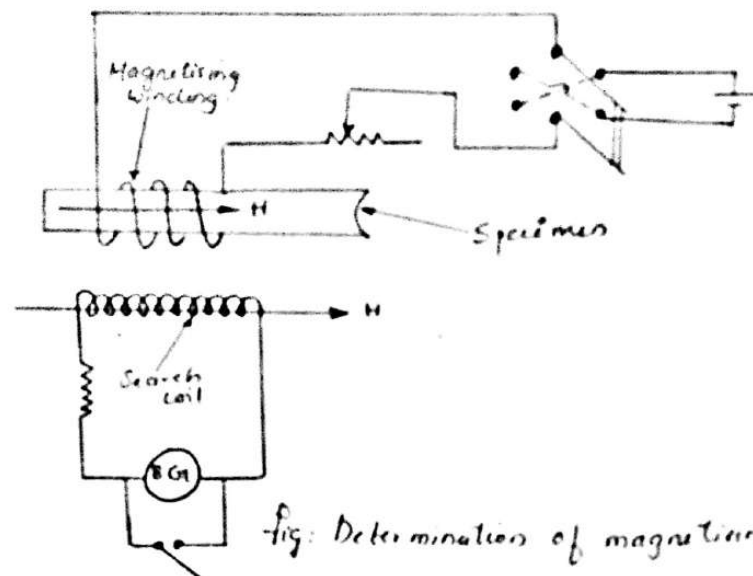


fig: Determination of magnetising force, H .

- >> The magnetising force (H) of a constant magnetic field can be measured by a ballistic galvanometer and search coil.
- >> The value of H cannot be measured directly.
- >> It can only be calculated from the flux density measured by the method of current reversal as in previous section.
- >> fig: shows the circuit diagram of the measurement of magnetising force H . The search coil used in the measurement of magnetising force is called "H coil".
- * In order to determine the magnetising force in the airgap, the H coil should be placed in the airgap itself.
- * But in order to determine the magnetising force of the ferromagnetic material, the coil H is placed in its surface since the tangential components of the field are equal in magnitude for both sides of the surface.
- >> If the current through the magnetising winding is reversed by using a reversal switch, the BG deflects, which is

proportional to the flux density B_0 .

⇒ The magnetising force (H) can be calculated using this method measured flux density B_0 , by using the formula.

$$H = \frac{B_0}{\mu_0} \text{ A/m} \quad \text{when } \mu_0 = 4\pi \times 10^{-7}$$

Determination of B-H curve.

There are 2 methods available for the determination of B-H curve of a specimen.

1. Method of Reversals.
2. Step by step Method.

1. Method of Reversals.

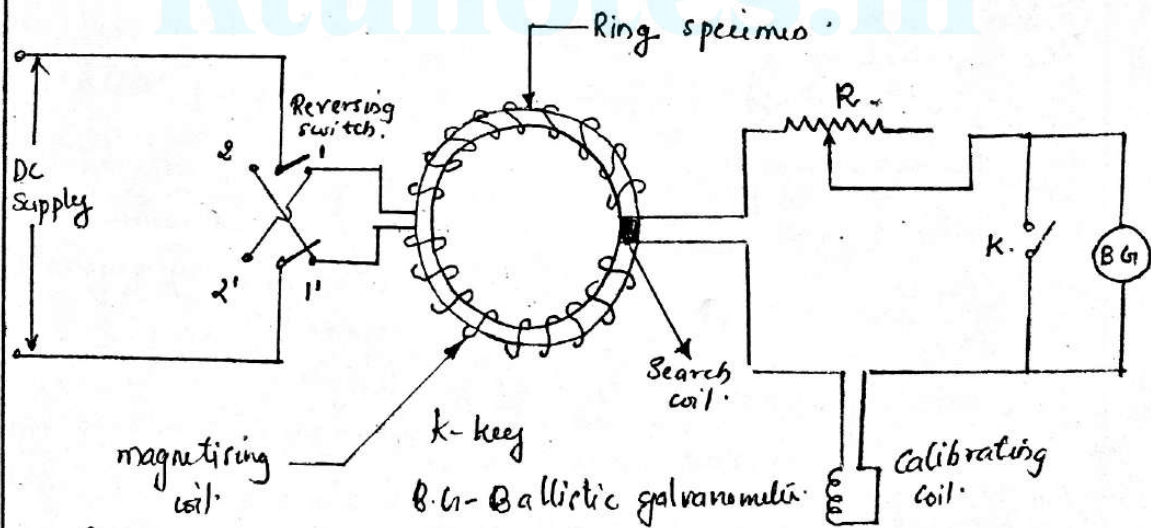


fig: Measurement of flux/flux density in ring specimens.

- For the determination of B-H curve, a ring shaped specimen whose dimensions are known is used for the purpose.
- A layer of thin tape is put on the ring and a search coil insulated by paraffined wax is wound over the tape. Another layer of tape is put over the search coil.

The magnetizing winding is uniformly wound over this tape. fig shows above.

Procedure:

1. 1st of all the specimen is demagnetised and then the magnetising current I is set to its lowest value.
2. Ballistic galvanometer key 'K' is closed & the reversing switch S is operated about 20 times backward & forward. This is done to bring the specimen into a reproducible cyclic magnetic state.
3. Key K is now opened and the value of flux corresponding to this value of H is measured by reversing the switch S and noting the throw of galvanometer.
4. The value of flux density corresponding to this H can be calculated by dividing the flux by area of the specimen.
5. The value of $H = \frac{NI_1}{l}$
 - N - no. of turns on the magnetising winding.
 - I_1 - corresponding magnetising current.
 - l - length
6. The above procedure is repeated for various values of H , by increasing H upto the maximum testing point value.

The graph of B against H may be plotted from the measured values of B corresponding to various values of H .

step by step method

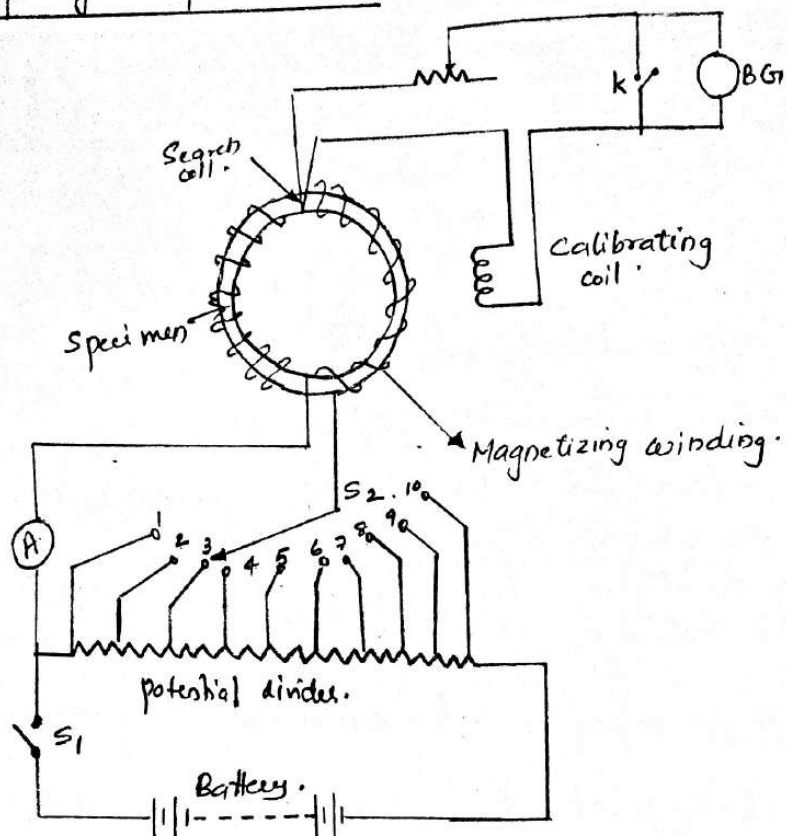


fig: Determination of B-H curve by step-by step method.

- 1) In this method reversal of magnetising current is not used. The magnetising current in the winding is supplied through a potential divider as shown in fig:
- 2) The potential divider has no. of tappings. The tappings are arranged in such a way that the magnetising force H may be increased in a no. of suitable steps upto the required max value.
- 3) The specimen before being tested is demagnetised.
- 4) The tapping switch S_2 is set on tapping 1 & the switch S_1 is closed. The throw of galvanometer corresponding to this increase in flux density in the specimen from 0 to B_1 . The value of B_1 can be calculated from the throw of the galvanometer. The value of corresponding magnetising

force H_1 may be calculated from the value of current flowing in the magnetising winding at tapping 1.

>> The magnetising force is then increased to H_2 by switching S_2 suddenly to tapping 2. and the corresponding increase in flux density ΔB is determined from the throw of the galvanometer. Then flux density B_2 corresponding to magnetising force H_2 can be obtained by $B_1 + \Delta B$.

>> This process is repeated for other values of H up to the maximum point of the complete $B-H$ curve is thus obtained as shown in fig below:

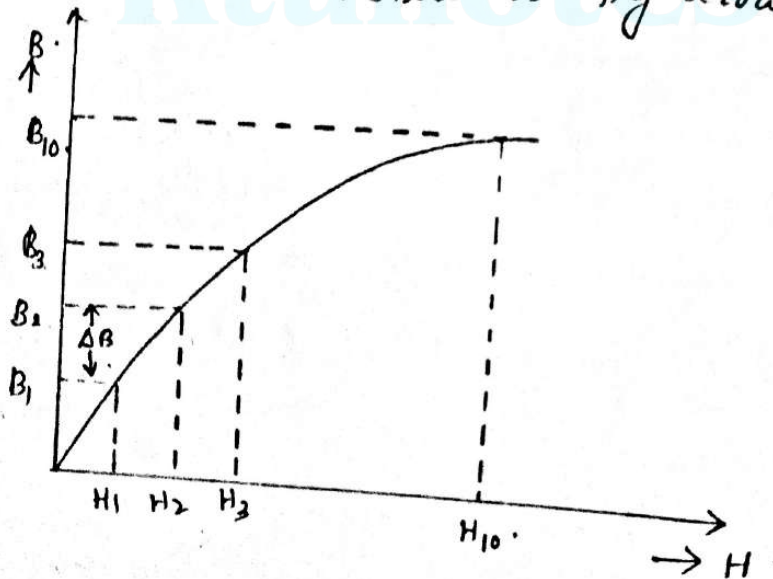


fig: $B-H$ curve from step-by-step method.